

Lab Class 10: Structural Bioinformatics

Joshua Robinson (PID: A18574862)

Table of contents

Introduction to the RCSB Protein Data Bank (PDB)	2
Visualizing the HIV-1 protease structure	3
Introduction to Bio3D in R	4
Comparative structure analysis of Adenylate Kinase	6
Search and retrieve ADK structures	6

```
pdb <- read.csv("pdb_stats (4).csv")
pdb
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	178795	21825	12773	343	226
2	Protein/Oligosaccharide	10363	3564	34	8	11
3	Protein/NA	9106	6335	287	24	7
4	Nucleic acid (only)	3132	221	1566	3	15
5	Other	175	25	33	4	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron Other Total					
1	84 32	214078				
2	1 0	13981				
3	0 0	15759				
4	3 1	4941				
5	0 0	237				
6	0 4	22				

```
pdb <- read.csv("pdb_stats (4).csv", check.names = FALSE, stringsAsFactors = FALSE)
```

Introduction to the RCSB Protein Data Bank (PDB)

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

About 81% for X-Ray and about 13% for Electron

```
total_structures <- sum(pdb$Total)
xray_total <- sum(pdb$`X-ray`)
em_total <- sum(pdb$EM)
xray_pct <- round(100 * xray_total / total_structures, 2)
em_pct <- round(100 * em_total / total_structures, 2)
xray_pct
```

```
[1] 80.95
```

```
em_pct
```

```
[1] 12.84
```

Q2: What proportion of structures in the PDB are protein?

About 97.9%.

```
total <- sum(pdb$Total)
protein <- pdb$Total[1] + pdb$Total[2] + pdb$Total[3]
protein_proportion <- protein / total
protein_proportion
```

```
[1] 0.979118
```

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are 1,227 HIV-1 protease structures in the current PDB.

Visualizing the HIV-1 protease structure



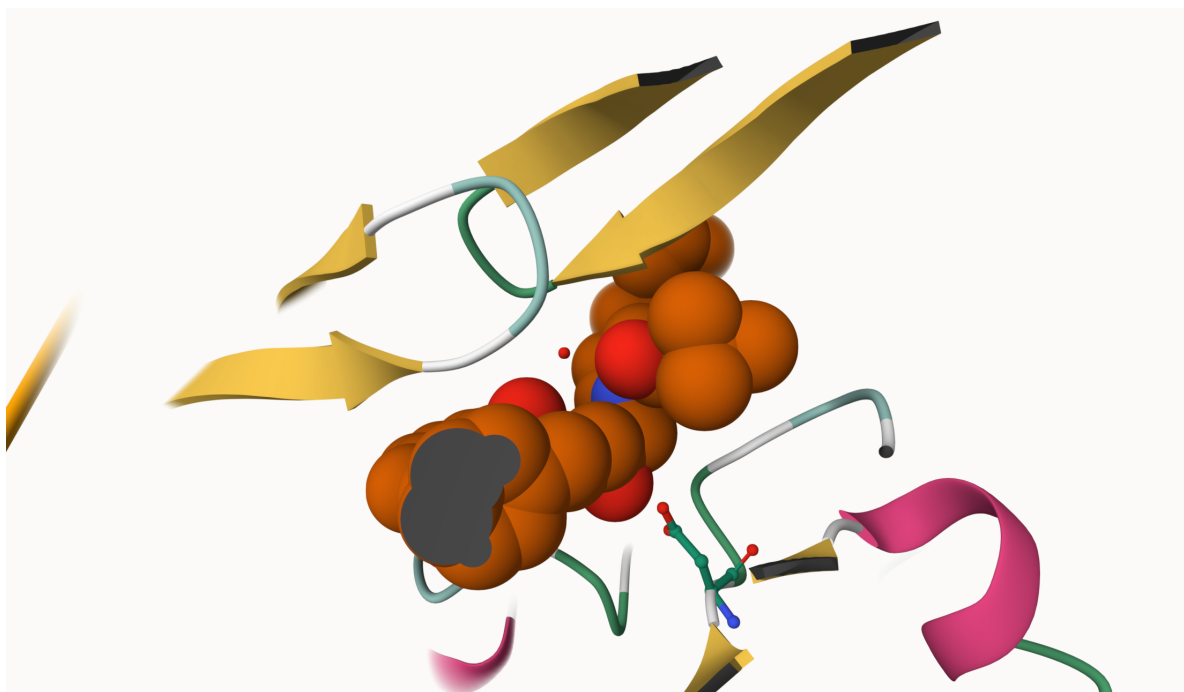
Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

Water molecules have 3 atoms, but in X-ray crystal structures the hydrogen atoms are usually not visible because they scatter X-rays very weakly. Therefore, only the oxygen atom is typically resolved and displayed in the PDB structure.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

HOH 308

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.



Introduction to Bio3D in R

```
library(bio3d)
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

There are 198 amino acid residues.

Q8: Name one of the two non-protein residues?

One of the two non-protein residues is HOH (water).

Q9: How many protein chains are in this structure?

There are 2 protein chains in this structure (chains A and B).

```
attributes(pdb)
```

```
$names
[1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"

$class
[1] "pdb" "sse"
```

```
head(pdb$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

Comparative structure analysis of Adenylate Kinase

Q10. Which of the packages above is found only on BioConductor and not CRAN?

msa

Q11. Which of the above packages is not found on BioConductor or CRAN?:

bio3dview

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

TRUE

Search and retrieve ADK structures

```
library(bio3d)
aa <- get.seq("lake_A")
```

Warning in get.seq("lake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

aa

```

      1      .      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
      1      .      .      .      .      .      .      60

      61      .      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPRTPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      .      120

     121      .      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
     121      .      .      .      .      .      .      180

     181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
     181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

+ attr: id, ali, call

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

This sequence contains 214 amino acids.